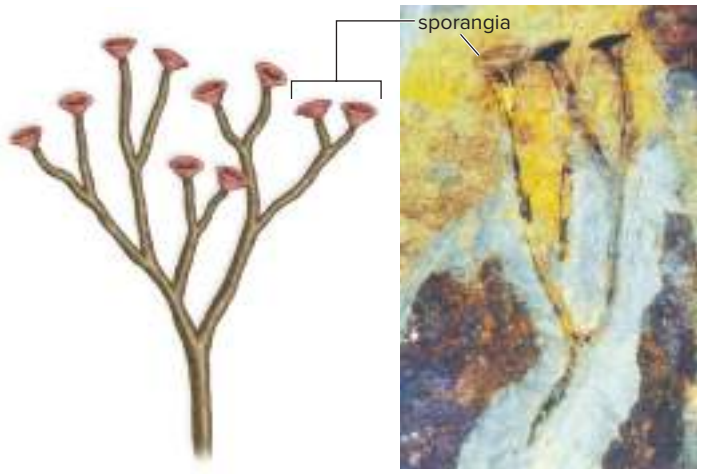


Figure 23.10 A *Cooksonia* fossil. The upright branches of a *Cooksonia* fossil, no more than a few centimeters tall, terminated in sporangia. (photo): Hans Steur, the Netherlands

The Origin of Vascular Plants

The fossil record indicates that the first vascular plants, such as *Cooksonia,* were more likely bushes than trees. *Cooksonia* is a rhyniophyte, a group of vascular plants that flourished during the Silurian period (about 416 MYA, see Table 18.1) but are now extinct. The rhyniophytes were only about 6.5 cm tall and had no roots or leaves. They consisted simply of a stem that forked evenly to produce branches ending in sporangia (Fig. 23.10).

The branching of *Cooksonia* was significant for two reasons. First, instead of the single sporangium in a bryophyte, this plant produced many sporangia, and therefore many more spores. Second, branching is characteristic of plants that have vascular tissue.

The sporangia of *Cooksonia* produced windblown spores, which classifies it as a **seedless vascular plant,** like the rest of the lycophytes and the pteridophytes, discussed in the next section. In addition, the lycophytes and all other vascular plants have a dominant sporophyte generation.

Lycophytes

Like the stem of early vascular plants, the first **lycophytes** also had leaves and roots. The leaves are called **microphylls,** because they had only one strand of vascular tissue. Microphylls most likely evolved as simple side extensions of the stem (see Fig. 23.12*a*). Roots evolved simply as lower extensions of the stem; the organization of vascular tissue in the roots of lycophytes today is much as it was in the stems of fossil vascular plants, where the vascular tissue is centrally placed.

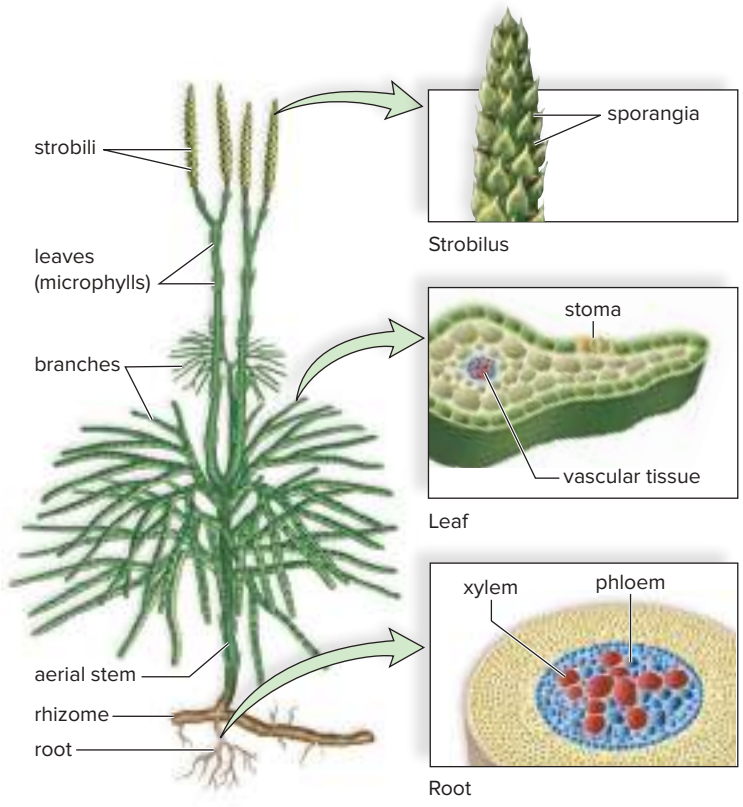
Today's lycophytes include around 1,300 species in three main groups: the ground pines or club mosses (*Lycopodium*), spike mosses (*Selaginella,* Fig. 23.11*a*), and quillworts (*Isoetes*). Many lycophytes have "moss" in their name because they look like a moss, yet unlike a moss, these plants have vascular tissue.

Figure 23.11*b* shows the structure of *Lycopodium;* beginning at the top, notice the cone-shaped structure called the **strobili** (*sing.,* strobilus [Gr. *strobilos,* "pinecone"]). The strobili resemble a club or heavy stick, accounting for the common name, club mosses. The sporangia are borne on the strobili, where meiosis takes place and spores are produced. Next, the aerial stem and microphylls contain vascular tissue, as does the underground stem, called the **rhizome.** The roots develop and branch from the rhizome.

Thus far, sporangia have been described as spore-producing structures. Some lycophytes and other vascular plants produce one type of spore that grows into one type of gametophyte. These plant types are called **homosporous** (see Fig. 23.16). Other vascular plants produce two types of spores. They are called **heterosporous.**



a.



b.

Figure 23.11 Lycophytes. **a.** *Selaginella, a* spike moss, is a common landscape or houseplant. **b.** *Lycopodium,* a club moss, has a sporophyte that develops an underground rhizome system. A rhizome is an underground stem. This rhizome produces true roots along its length. (a): Steven P. Lynch

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The two types of spores are microspores and megaspores. **Microspores** grow into a male gametophyte, and **megaspores** grow into a female gametophyte (see Fig. 23.18).

Check Your Progress 23.3

1. Name two features of lycophytes that increase their ability to survive on land.
2. Explain how xylem contributes to an upright body plant.
3. Define the terms *homosporous* and *heterosporous*.

23.4 Evolution of Pteridophytes: Megaphylls

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify three types of pteridophytes.
2. Compare and contrast microphylls and megaphylls.
3. Identify the components of the ferns' life cycle.

Pteridophytes is a broad term used to describe a group of seedless vascular plants that includes ferns and their allies, the horsetails and whisk ferns. Both pteridophytes and the seed plants have megaphylls. **Megaphylls** are broad leaves with several strands of vascular tissue. Figure 23.12*a* shows the difference between microphylls and megaphylls, and Figure 23.12*b* shows how megaphylls could have evolved.

Megaphylls, which evolved independently from the evolution of microphylls, allow plants to efficiently collect solar energy, leading to the production of more food and the possibility of producing more offspring than plants without megaphylls. Therefore, the evolution of megaphylls made plants more fit. Recall that fitness, in an evolutionary sense, is judged by the number of living offspring an organism produces relative to others of its own kind.

The pteridophytes, like the lycophytes, were dominant from the late Devonian period through the Carboniferous period. Today, the lycophytes are quite small, but some of the extinct relatives of today's club mosses were 35 m tall and dominated the Carboniferous swamps. The horsetails (around 18 m tall) and ancient tree ferns (8 m or more in height) also contributed significantly to the great swamp forests of the time (see the following Evolution feature, "Carboniferous Forests").

Horsetails

Today, **horsetails** consist of one genus, *Equisetum,* and approximately 15 species of distinct seedless vascular plants. Most horsetails inhabit wet, marshy environments around the globe. About 299 MYA, horsetails were dominant plants and grew as large as

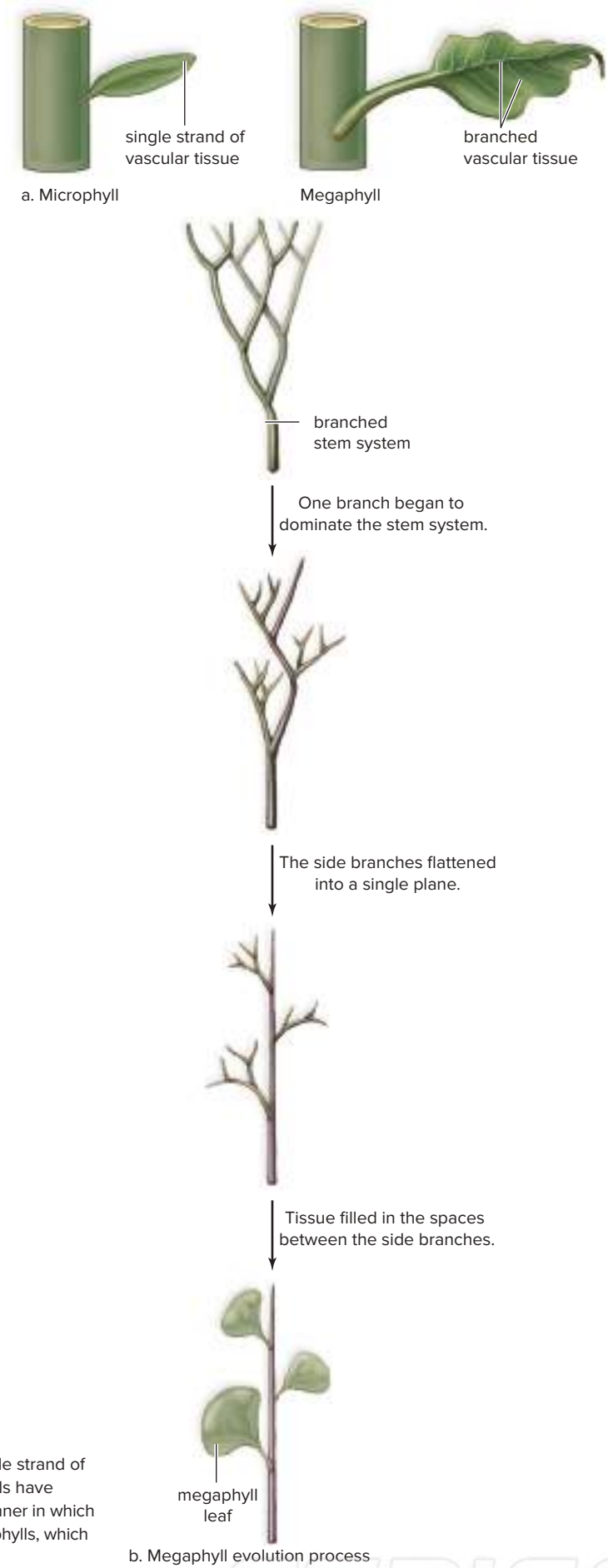


Figure 23.12 Microphylls and megaphylls. **a.** Microphylls have a single strand of vascular tissue, which explains why they are quite narrow. In contrast, megaphylls have several branches of vascular tissue and are broader. **b.** These steps show the manner in which megaphylls may have evolved. All vascular plants, except lycophytes, bear megaphylls, which can gather more sunlight and produce more organic food than microphylls.

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